



THE IMPACT OF INTERNET USAGE ON POLITICAL POLARIZATION IN THE U.S.

A DATA-DRIVEN INVESTIGATION

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WHAT IS POLITICAL POLARIZATION?

Polarization - division into two sharply distinct opposites

especially; a state in which the opinions, beliefs, or interests of a group or society no longer range along a continuum but become concentrated at opposing extremes

(Merriam-Webster)

Hence, we will be investigating the impact of internet use on political polarization in the U.S., defined as the growing divide between partisan affiliations and ideologies.

PROBLEM SIGNIFICANCE

- Target Client
 - any group/body trying to better understand and/or mitigate the societal impact that internet usage has on political polarization: internet users, voters, media companies, government and policymakers.
- Why it Matters
 - Political polarization
 - undermines democratic discourse,
 - fuels social tension and
 - increases legislative gridlock
- Investigating the effect of internet usage will give us perspective and help to explain if it is shaping political attitudes in society today.

FACTS/STATISTICS

- Today, 72% of Republicans regard Democrats as more immoral, and 63% of Democrats say the same about Republicans. (Pew Research, 2022).



- A large majority of U.S. adults (86%) say they at least sometimes get news from a smartphone, computer or tablet, including 57% who say they do so often. (Pew Research, 2024)

RESEARCH QUESTION(S)

- Does more frequent internet use increase the likelihood of strong partisan affiliation among US adults?
- Does the effect of internet use on partisan affiliation vary by age or race?

PRIOR WORKS

Source	Research Question	Predictors	Methods	Findings
Sounman Hong & Sun Hyoung Kim (2016) <i>Political Polarization on Twitter: Implications for the Use of Social Media in Digital Governments</i>	Does social media, especially Twitter, make political polarization worse by giving a platform to extreme voices?	DW-Nominate score, age, education, religion, social media usage, traditional media exposure, political tenure, activity on Twitter, and length of Twitter usage.	Regression models to examine the relationship between ideological extremity and number of followers on Twitter	Politicians with extreme ideological views (liberal or conservative) had significantly more Twitter followers than moderate politicians.
Bail, Christopher A., et al. (2018) <i>Exposure to Opposing Views on Social Media Can Increase Political Polarization.</i>	Does exposure to opposing political views on social media platforms increase political polarization?	Partisan identity, Twitter engagement level, Initial political orientation, Exposure to opposing-view bot, Engagement with bot's content	Multiple linear regressions estimating the effect of being assigned to follow the Twitter bot and effect on those who followed the bot and engaged with the tweets.	Republicans who followed a liberal Twitter bot became significantly more conservative, while Democrats who followed a conservative bot became slightly more liberal.

Source	Research Question	Predictors	Methods	Findings
Bryson, B. P. (2020). <i>Polarizing the middle: internet exposure and public opinion</i> .	Does internet exposure polarize the middle (moderates, independents)? Whether respondents who likely accessed political info online during survey exhibit greater polarization than those whose political knowledge was retained?	Bachelor's degree (y/n), income, age, race, gender, internet access (y/n), degree of political interest, web survey (control variable), high knowledge, cheaters	Quantile regression to better capture the effect of predictors on the entire distribution of the dependent variable (ideological consistency).	The moderate and less engaged groups are easily polarized when have access to online information. Those without a bachelor's degree and those with less interest in politics were primarily pulled to the right. Independents were pulled to the left. Moderates were pulled in both directions.
Ludwig, K., Grote, A., Iana, A., Alam, M., Paulheim, H., Sack, H., ... & Müller, P. (2023). Divided by the algorithm? the (limited) effects of content- and sentiment-based news recommendation on affective, ideological, and perceived polarization.	How content-based news recommendation systems (NRS) influence affective, ideological, and perceived polarization?	plain content-based NRS, negative plain content-based NRS, positive plain content-based NRS, random article selection, time spent, age, gender, education, political interest...	Linear regression. Model 1 includes NRS variables and time spent. Model 2 includes interaction terms between NRSs and time spent. Control variables like age, gender... are in both models.	limited evidence for polarization effects of content-based NRS in general compared to random news articles. The longer participants used an NRS enriched with negative sentiment, the more they became affectively polarized.

Source	Research Question	Predictors	Methods	Findings
Boxell et.al. (2017) <i>Greater Internet use is not associated with faster growth in political polarization among US demographic groups</i>	Does greater internet use lead to faster growth in political polarization among U.S. demographic groups?	Internet use (yes/no), age, education, race, gender, income, region	Regression Analysis and an 8-measure polarization index Data: American National Election Studies (1996 – 2012)	Internet use grew from 36% to 73% (1996–2012) Polarization increased more among low-use groups (e.g., 65+: 0.6–1.0 points on index) than high-use groups (e.g., under 35: 0.2–0.4 points); internet use not a primary driver of polarization growth.
Melnikov (2023) <i>Mobile Internet and Political Polarization</i>	How has mobile internet affected political polarization in the U.S. and what mechanisms drive this effect?	3G network coverage, political affiliation, misinformation and traditional news consumption	Difference in differences analysis Data: Gallup Daily Poll data (2008 – 2017)	3G access increased polarization, with Republicans becoming more conservative and Democrats more liberal; (11.3% of polarization increase explained)

DATA SOURCE

- The American Trends Panel, from Pew Research Center, from 2020-2023
- <https://www.pewresearch.org/the-american-trends-panel/>

INTERNET Factor Internet usage frequency

In 2021–2023: 1 = Almost constantly; 2 = Several times a day; 3 = About once a day; 4 = Several times a week; 5 = Less often; 6 = Rarely or never. In 2020: 1 = Frequent user; 2 = Infrequent user

RACE Factor Race of respondent

1 = White, non-Hispanic; 2 = Black, non-Hispanic; 3 = Hispanic; 4 = Asian, non-Hispanic; 5 = Other / Multiple races

AGE Numeric Age category of respondent (numericalized)

1 = 18–29 years old; 2 = 30–49 years old; 3 = 50–64 years old; 4 = 65+ years old

GENDER Factor Gender of respondent

1 = Male; 2 = Female

EDU Factor Educational attainment

1 = High school or less; 2 = Some college, no degree; 3 = Associate degree; 4 = Bachelor's degree; 5 = Postgraduate degree

INCOME Numeric Income group (numericalized)

Low income, middle income, high income.

Dependent
variable

REP Factor Favorability toward the Republican Party

1 = Very unfavorable; 2 = Mostly unfavorable; 3 = Mostly favorable; 4 = Very favorable

Dependent
variable

DEM Factor Favorability toward the Democratic Party

1 = Very unfavorable; 2 = Mostly unfavorable; 3 = Mostly favorable; 4 = Very favorable

PARTY Factor Political party identification

1 = Republican; 2 = Democrat

PREDICTOR TABLE

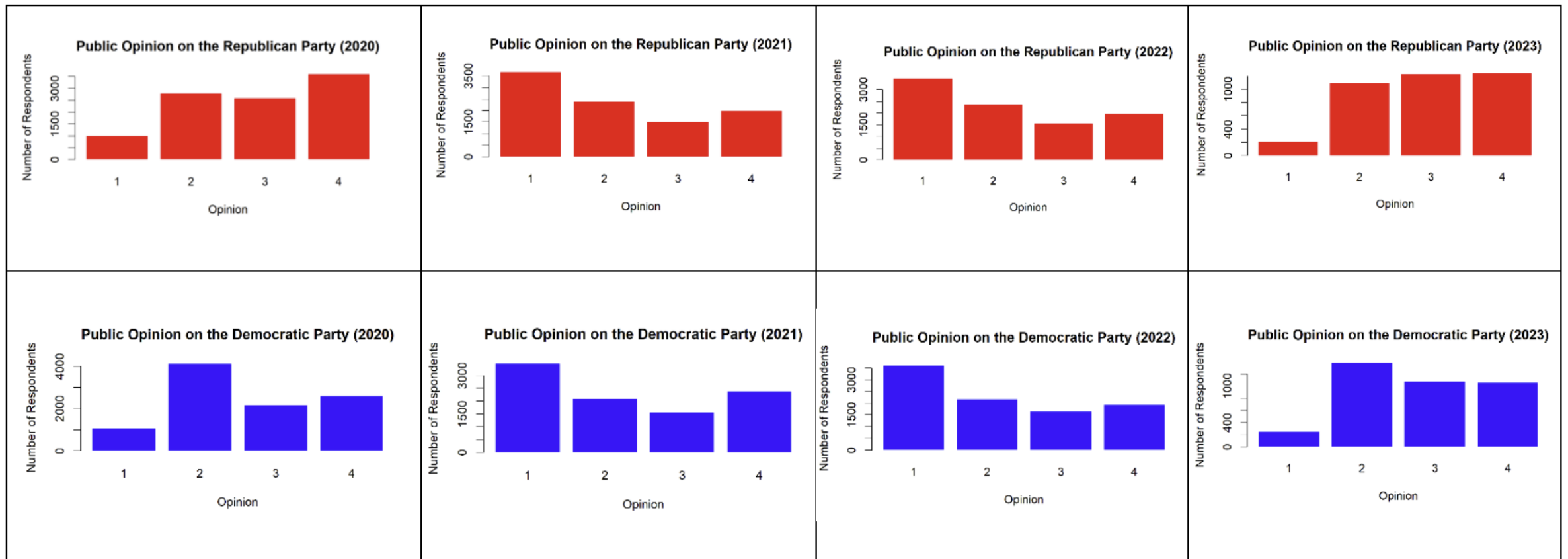
Variable	Effect	Rationale
INTERNET	+/-	Internet usage may exacerbate political polarization
PARTY	+/-	Political party identification, for possible interaction terms
AGE	+/-	Contextual variable for age
RACE	+/-	Contextual variable for race
GENDER	+/-	Contextual variable for gender
EDU	+/-	Contextual variable for education level
INCOME	+/-	Contextual variable for household income

DATA PREPARATION

- Removed rows with missing values, “unknown” or “refused to answer” responds
- DVs of 2020 range from 0-100, while 2021, 2022, 2023 range from 1 – 4 of degree of favorability. Rescaled 2020 DVs to be consistent with the other years by assigning 1 – 4 based on quartiles.
- Drop irrelevant variables from data frames and renamed columns for easier interpretation.
- Approximated ‘age’ into median values of each age category. 77 is the average age of death in the US and is used as the upper limit for the last age category.
- Approximated ‘income’ into median values of each income category. \$200,000 is assumed as the upper limit for the last category.
- For ‘gender’, only include male and female in this analysis.
- Converted categorical variables to factors.

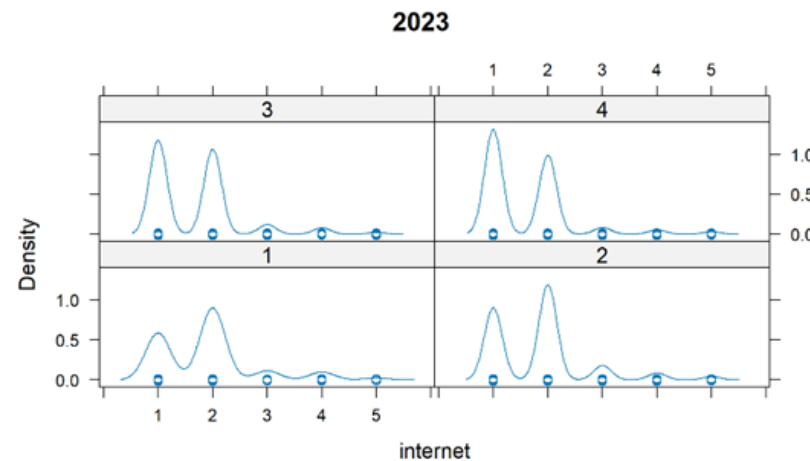
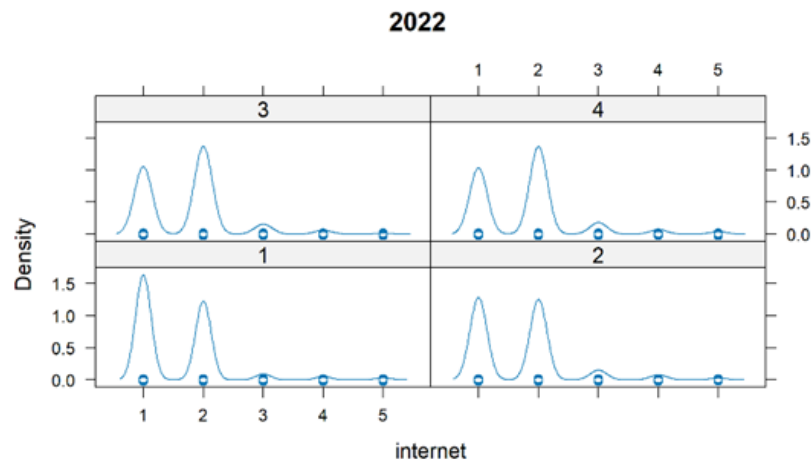
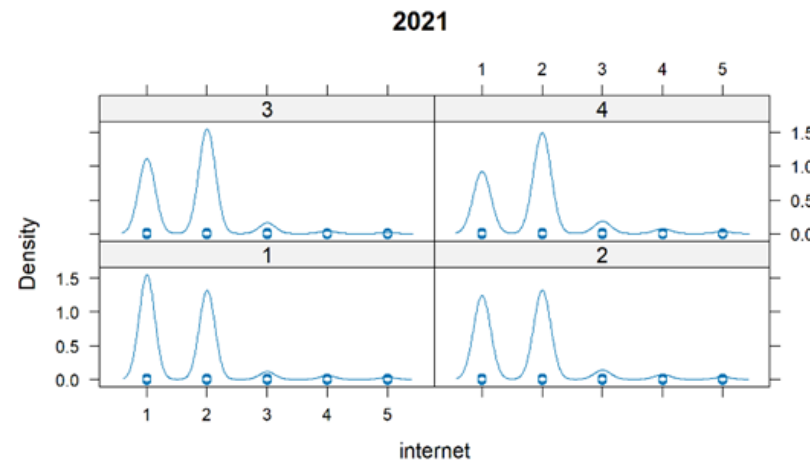
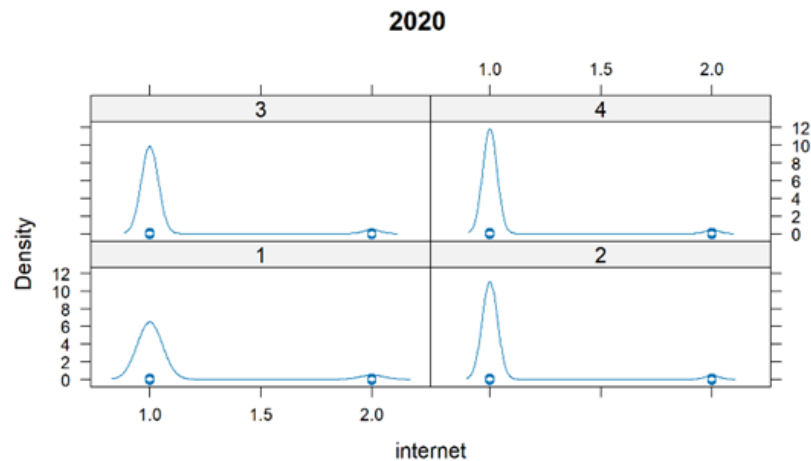
DATA VISUALIZATION

■ Distribution of the DVs



DATA VISUALIZATION

■ Internet usage across different levels of favorability towards **republicans**



Internet Usage

1 = Almost constantly;
2 = Several times a day;
3 = About once a day;
4 = Several times a week;
5 = Less often; Rarely or never.

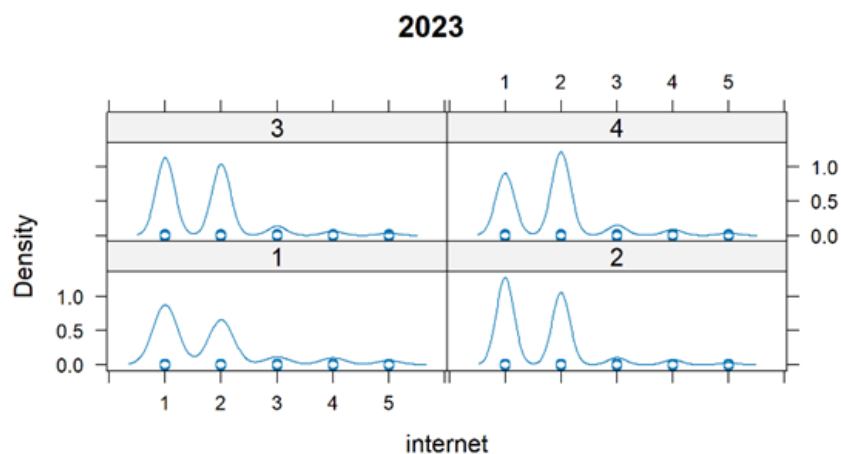
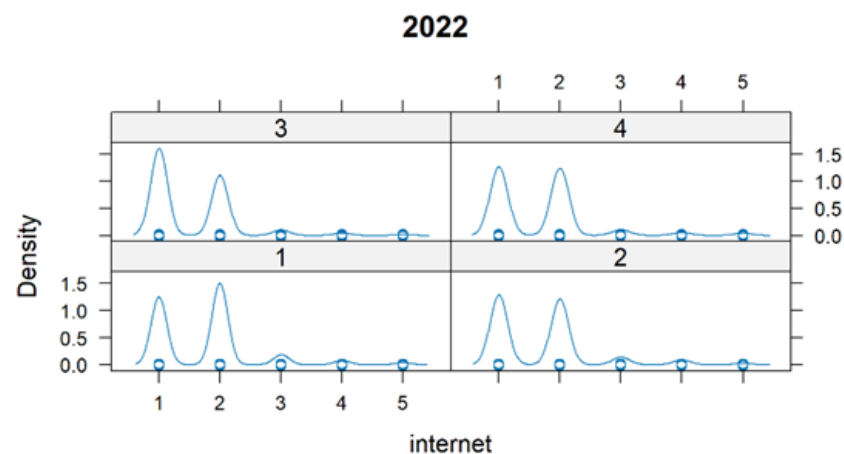
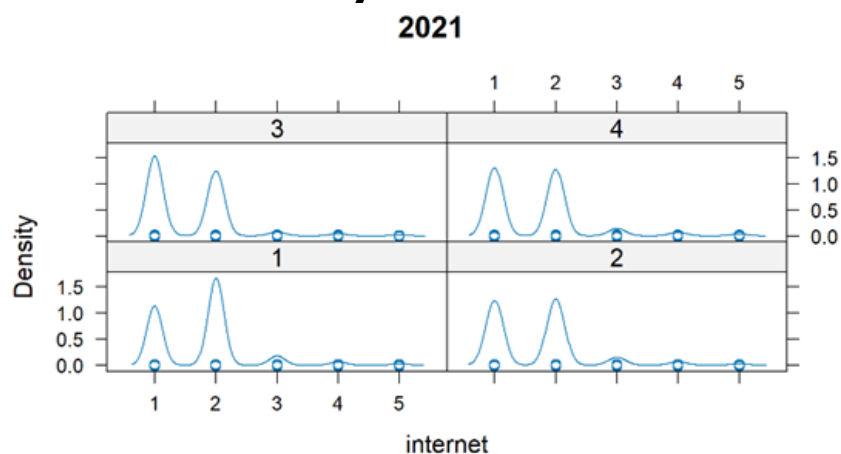
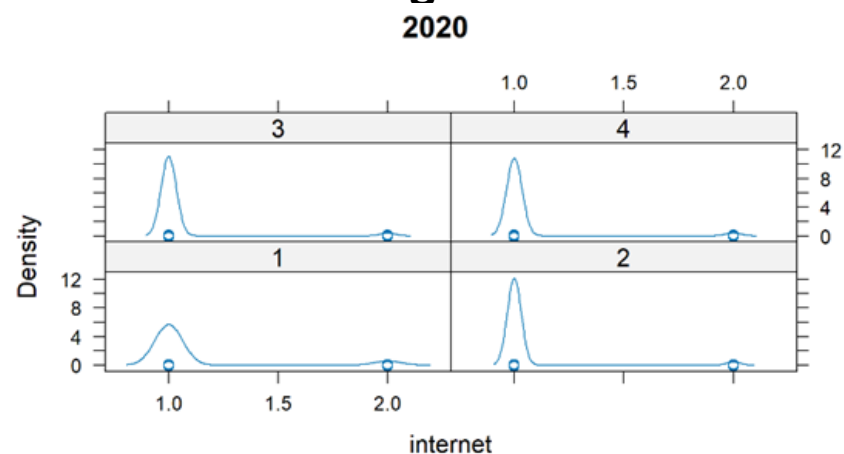
In 2020: 1 = Frequent user; 2 = Infrequent user

Levels of favorability

1 = Very unfavorable;
2 = Mostly unfavorable;
3 = Mostly favorable;
4 = Very favorable

DATA VISUALIZATION

■ Internet usage across different levels of favorability towards democrats



Internet Usage

1 = Almost constantly;
2 = Several times a day;
3 = About once a day;
4 = Several times a week;
5 = Less often; Rarely or never.

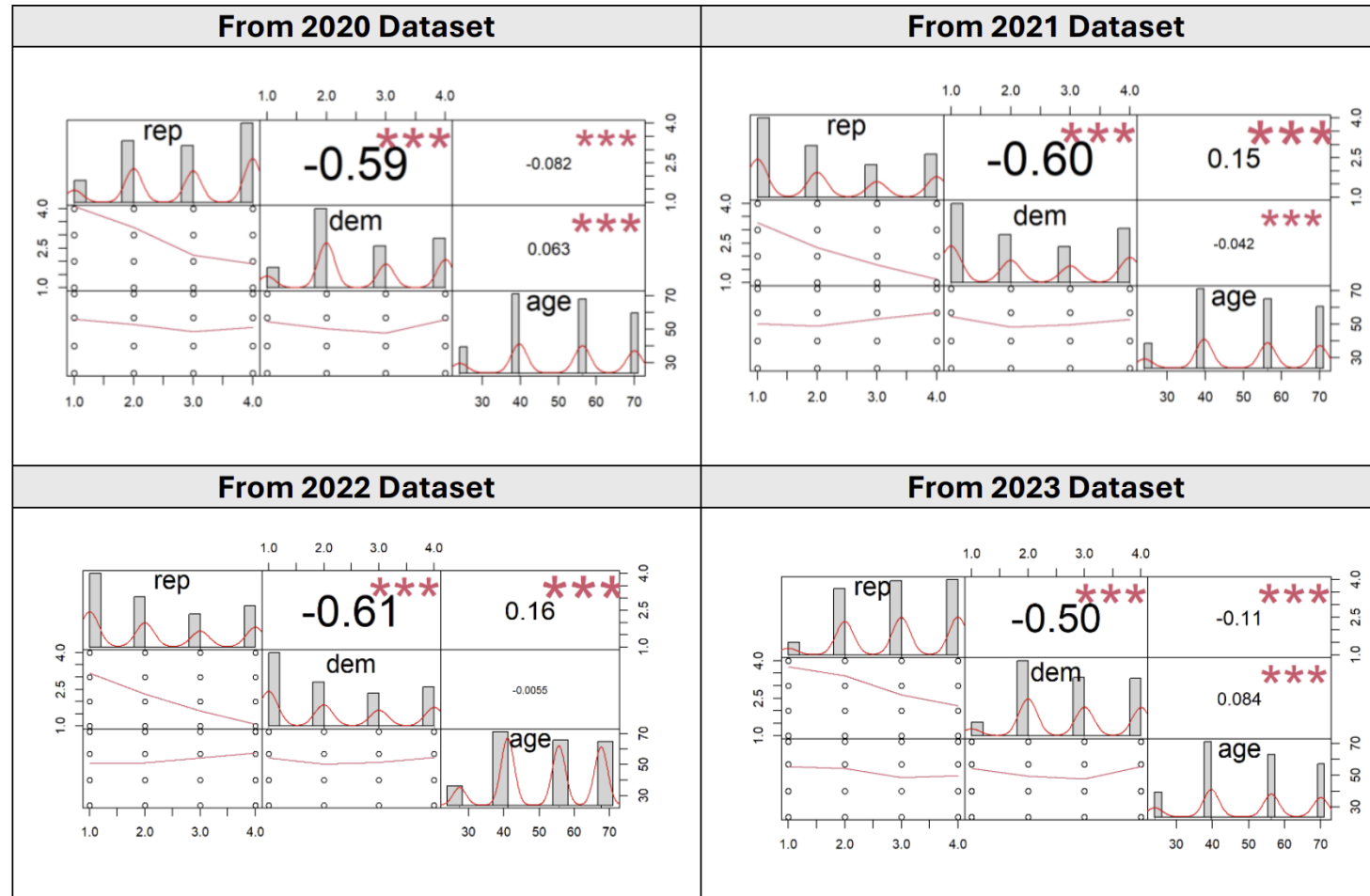
In 2020: 1 = Frequent user; 2 = Infrequent user

Levels of favorability

1 = Very unfavorable;
2 = Mostly unfavorable;
3 = Mostly favorable;
4 = Very favorable

MODELS

■ Correlation Analysis



MODELS DESCRIPTION

Ordinal Logistic Regression (OLR)

The dependent variables **rep** and **dem** measure how favorable or unfavorable a respondent is toward each party on a 1-4 ordinal scale. Ordinal Logistic Regression (OLR) is the appropriate modeling technique.

We will use a :

Base Model: To examine the main effect of **internet usage** on the likelihood of a respondent rating their favorability toward a party, adjusting for factors like **age**, **gender**, **income**, and **education**.

This will address:

RQ1: Does more frequent internet use increase the likelihood of strong partisan affiliation among U.S. adults?

Model with Interaction Terms: To explore whether the **effect of internet usage** on political affiliation is stronger or weaker across **age**, **race**, and **party**.

This will address:

RQ2: Does the effect of internet use on partisan affiliation vary by age or race?

MODELS & OUTPUT

Models for **rep** dependent variable

- model_rep_2020 <- polr(rep ~ internet + age + gender + edu + race + party + log(income), data = df2020, Hess = TRUE)
- model_rep_2021 <- polr(rep ~ internet + age + gender + edu + race + party + log(income), data = df2021, Hess = TRUE)
- model_rep_2022 <- polr(rep ~ internet + age + gender + edu + race + party + log(income), data = df2022, Hess = TRUE)
- model_rep_2023 <- polr(rep ~ internet + age + gender + edu + race + party + log(income), data = df2023, Hess = TRUE)

Model for **dem** dependent variable

- model_dem_2020 <- polr(dem ~ internet + age + gender + edu + race + party + log(income), data = df2020, Hess = TRUE)
- model_dem_2021 <- polr(dem ~ internet + age + gender + edu + race + party + log(income), data = df2021, Hess = TRUE)
- model_dem_2022 <- polr(dem ~ internet + age + gender + edu + race + party + log(income), data = df2022, Hess = TRUE)
- model_dem_2023 <- polr(dem ~ internet + age + gender + edu + race + party + log(income), data = df2023, Hess = TRUE)

Effect of Internet Use on Republican Affiliation

	Dependent variable:			
	rep			
	2020 (1)	2021 (2)	2022 (3)	2023 (4)
internet2	-0.225** (0.102)	0.108** (0.046)	0.018 (0.046)	-0.168** (0.072)
internet3		0.135 (0.099)	0.242** (0.098)	-0.423*** (0.153)
internet4		0.277* (0.146)	0.057 (0.137)	-0.566*** (0.191)
internet5		0.125 (0.213)	0.140 (0.202)	-0.554* (0.302)
age	-0.003* (0.001)	0.005*** (0.001)	0.009*** (0.002)	-0.0004 (0.002)
gender2	-0.256*** (0.041)	0.200*** (0.041)	0.209*** (0.042)	-0.198*** (0.066)
edu2	-0.313*** (0.047)	0.191*** (0.047)	0.227*** (0.048)	-0.293*** (0.077)
edu3	-0.601*** (0.064)	0.347*** (0.062)	0.399*** (0.062)	-0.486*** (0.093)
race2	-0.082 (0.076)	-0.038 (0.080)	0.179** (0.079)	-0.026 (0.095)
race3	-0.334*** (0.119)	0.064 (0.115)	0.298*** (0.108)	-0.368*** (0.118)
race4	0.171 (0.105)	-0.243** (0.107)	-0.116 (0.110)	0.026 (0.155)
race5	-0.171 (0.105)	0.113 (0.111)	0.459*** (0.116)	-0.320* (0.172)
party2	3.904*** (0.059)	-3.538*** (0.057)	-3.718*** (0.059)	3.213*** (0.088)
log(income)	0.174*** (0.025)	-0.040 (0.029)	0.006 (0.030)	0.084* (0.048)
Observations	9,999	9,542	9,374	3,778

Note: *p<0.1; **p<0.05; ***p<0.01

Effect of Internet Use on Democratic Affiliation

	Dependent variable:			
	dem			
	2020 (1)	2021 (2)	2022 (3)	2023 (4)
internet2	-0.321*** (0.106)	-0.062 (0.047)	0.066 (0.047)	0.014 (0.073)
internet3		0.029 (0.104)	-0.166 (0.106)	-0.031 (0.157)
internet4		-0.130 (0.150)	-0.254* (0.150)	-0.081 (0.201)
internet5		-0.392* (0.224)	-0.231 (0.219)	0.169 (0.314)
age	-0.003** (0.001)	0.013*** (0.002)	0.018*** (0.002)	-0.003 (0.002)
gender2	-0.317*** (0.041)	0.262*** (0.042)	0.241*** (0.042)	-0.288*** (0.066)
edu2	0.217*** (0.048)	-0.215*** (0.049)	-0.282*** (0.050)	0.187*** (0.079)
edu3	0.041 (0.066)	-0.265*** (0.064)	-0.393*** (0.066)	0.103 (0.095)
race2	-0.648*** (0.078)	0.368*** (0.077)	0.185** (0.077)	-0.277*** (0.097)
race3	-0.120 (0.121)	0.271** (0.108)	0.251** (0.103)	-0.513*** (0.119)
race4	-0.034 (0.106)	0.068 (0.106)	-0.155 (0.111)	0.060 (0.153)
race5	-0.588*** (0.108)	0.147 (0.110)	0.015 (0.118)	-0.470*** (0.176)
party2	-3.760*** (0.058)	3.937*** (0.061)	3.834*** (0.060)	-3.446*** (0.093)
log(income)	0.112*** (0.026)	0.019 (0.029)	0.081*** (0.030)	0.039 (0.048)
Observations	9,999	9,542	9,374	3,778

Note: *p<0.1; **p<0.05; ***p<0.01

MODELS & OUTPUT

Model for **rep** with interaction terms

- **modell_rep_2020** <- polr(rep ~ + internet * race + internet * party + gender + edu + log(income), data = df2020, Hess = TRUE)
- **modell_rep_2021** <- polr(rep ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2021, Hess = TRUE)
- **modell_rep_2022** <- polr(rep ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2022, Hess = TRUE)
- **modell_rep_2023** <- polr(rep ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2023, Hess = TRUE)

Interaction Effects of Internet Use on Republican Affiliation

	Dependent variable:			
	2020 (1)	2021 (2)	2022 (3)	2023 (4)
internet2	0.054 (0.434)	0.217 (0.172)	0.090 (0.178)	-0.699*** (0.271)
internet3		0.259 (0.545)	-0.172 (0.546)	-1.480** (0.708)
internet4		1.565* (0.872)	-0.334 (0.763)	-0.581 (1.010)
internet5		0.778 (1.210)	0.498 (1.257)	2.305 (1.657)
age	-0.002* (0.001)	0.006*** (0.002)	0.009*** (0.002)	-0.006* (0.004)
race2	-0.122 (0.080)	-0.110 (0.110)	0.129 (0.107)	0.033 (0.131)
race3	-0.288** (0.122)	0.108 (0.134)	0.364*** (0.124)	-0.336** (0.145)
race4	0.126 (0.107)	-0.242 (0.150)	-0.021 (0.152)	0.224 (0.221)
race5	-0.139 (0.109)	0.107 (0.141)	0.350** (0.158)	-0.342 (0.224)
party2	3.909*** (0.060)	-3.405*** (0.075)	-3.726*** (0.076)	3.141*** (0.116)
gender2	-0.255*** (0.041)	0.207*** (0.042)	0.214*** (0.042)	-0.193*** (0.066)
edu2	-0.310*** (0.047)	0.189*** (0.048)	0.225*** (0.048)	-0.294*** (0.078)
edu3	-0.607*** (0.064)	0.345*** (0.063)	0.399*** (0.063)	-0.482*** (0.094)
log(income)	0.174*** (0.025)	-0.040 (0.029)	0.003 (0.030)	0.093* (0.048)
internet2:age	-0.005 (0.007)	-0.0001 (0.003)	-0.001 (0.003)	0.010** (0.005)
internet3:age		-0.001 (0.008)	0.005 (0.008)	0.019* (0.011)
internet4:age		-0.020 (0.013)	0.002 (0.012)	-0.002 (0.015)
internet5:age		-0.007 (0.018)	-0.004 (0.019)	-0.034 (0.026)
internet2:race2	0.421 (0.271)	0.166 (0.170)	0.189 (0.170)	-0.193 (0.201)
internet3:race2		0.586 (0.405)	-0.436 (0.413)	0.155 (0.443)
internet4:race2		-0.309 (0.426)	0.194 (0.375)	1.147** (0.492)
internet5:race2		-0.599 (0.607)	-0.358 (0.601)	-1.946** (0.789)
internet2:race3	-1.032* (0.576)	-0.186 (0.278)	-0.476 (0.295)	-0.120 (0.260)
internet3:race3		0.212 (0.777)	-0.418 (0.556)	1.076 (0.779)
internet4:race3		-2.734* (1.491)	1.833** (0.887)	0.605 (0.961)
internet5:race3		0.012 (1.060)	0.024 (1.314)	-5.240** (2.058)
internet2:race4	1.143** (0.583)	0.014 (0.222)	-0.360 (0.237)	-0.533 (0.329)
internet3:race4		-0.085 (0.576)	0.072 (0.452)	0.425 (0.727)
internet4:race4		-0.862 (0.702)	0.741 (0.576)	0.736 (0.873)
internet5:race4		0.821 (1.113)	11.596*** (0.00000)	-1.516 (1.875)
internet2:race5	-0.404 (0.401)	0.052 (0.246)	0.243 (0.248)	0.166 (0.394)
internet3:race5		-0.598 (0.540)	0.371 (0.516)	0.291 (0.745)
internet4:race5		0.314 (0.604)	-0.114 (0.941)	-0.174 (0.834)
internet5:race5		-1.319 (1.118)	0.035 (0.789)	-0.780 (1.261)
internet2:party2	-0.085 (0.221)	-0.258*** (0.089)	-0.042 (0.090)	0.226 (0.145)
internet3:party2		-0.278 (0.205)	0.289 (0.202)	-0.272 (0.320)
internet4:party2		0.084 (0.333)	0.278 (0.293)	-0.410 (0.454)
internet5:party2		-0.012 (0.523)	-0.075 (0.466)	0.075 (0.727)
Observations	9,999	9,542	9,374	3,778

Note:

*p<0.1; **p<0.05; ***p<0.01

MODELS & OUTPUT

Model for "dem" with interaction terms

- **modell_dem_2020** <- polr(dem ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2020, Hess = TRUE)
- **modell_dem_2021** <- polr(dem ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2021, Hess = TRUE)
- **modell_dem_2022** <- polr(dem ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2022, Hess = TRUE)
- **modell_dem_2023** <- polr(dem ~ internet * age + internet * race + internet * party + gender + edu + log(income), data = df2023, Hess = TRUE)

Interaction Effects of Internet Use on Democratic Affiliation

	Dependent variable:			
	2020 (1)	2021 (2)	2022 (3)	2023 (4)
internet2	-0.330 (0.451)	-0.556*** (0.179)	-0.211 (0.190)	0.670** (0.275)
internet3		-0.863 (0.600)	-1.515** (0.648)	1.927** (0.749)
internet4		-0.483 (0.911)	-1.626* (0.856)	-0.479 (0.997)
internet5		0.834 (1.249)	-1.739 (1.459)	1.807 (1.690)
age	-0.003** (0.001)	0.010*** (0.002)	0.016*** (0.002)	0.002 (0.004)
race2	-0.535*** (0.082)	0.351*** (0.103)	0.158 (0.101)	-0.169 (0.134)
race3	-0.132 (0.124)	0.349*** (0.124)	0.345*** (0.116)	-0.602*** (0.144)
race4	-0.041 (0.108)	0.169 (0.142)	-0.193 (0.148)	-0.138 (0.210)
race5	-0.550*** (0.112)	0.270* (0.139)	0.224 (0.159)	-0.625*** (0.231)
party2	-3.771*** (0.059)	3.655*** (0.077)	3.624*** (0.077)	-3.163*** (0.118)
gender2	-0.320*** (0.041)	0.252*** (0.042)	0.233*** (0.042)	-0.290*** (0.067)
edu2	0.212*** (0.048)	-0.220*** (0.049)	-0.280*** (0.050)	0.193** (0.079)
edu3	0.054 (0.066)	-0.262*** (0.065)	-0.374*** (0.066)	0.097 (0.095)
log(income)	0.111*** (0.026)	0.023 (0.030)	0.085*** (0.030)	0.039 (0.048)
internet2:age	0.002 (0.007)	0.004 (0.003)	0.0005 (0.003)	-0.008 (0.005)
internet3:age		0.011 (0.009)	0.019** (0.010)	-0.030*** (0.011)
internet4:age		0.003 (0.013)	0.026** (0.013)	0.011 (0.015)
internet5:age		-0.025 (0.019)	0.020 (0.021)	-0.027 (0.025)
internet2:race2	-1.305*** (0.277)	0.143 (0.164)	0.042 (0.166)	-0.138 (0.205)
internet3:race2		-1.079*** (0.415)	0.302 (0.414)	-0.825* (0.463)
internet4:race2		0.164 (0.429)	0.714* (0.374)	-0.673 (0.538)
internet5:race2		-0.328 (0.616)	-0.567 (0.635)	0.242 (0.818)
internet2:race3	0.215 (0.597)	-0.043 (0.260)	-0.479* (0.281)	0.125 (0.267)
internet3:race3		-1.508* (0.844)	0.250 (0.602)	0.484 (0.805)
internet4:race3		-2.430* (1.331)	0.195 (0.772)	1.934* (1.171)
internet5:race3		-2.918 (1.997)	-11.250*** (0.00000)	0.526 (1.500)
internet2:race4	0.063 (0.519)	-0.065 (0.223)	0.213 (0.239)	0.330 (0.323)
internet3:race4		-0.975 (0.626)	-0.337 (0.514)	1.333* (0.737)
internet4:race4		-0.567 (0.600)	0.153 (0.641)	-1.238 (1.033)
internet5:race4		-2.919** (1.193)	-11.116*** (0.00000)	9.936*** (0.0001)
internet2:race5	-0.805* (0.447)	-0.224 (0.245)	-0.291 (0.249)	0.331 (0.393)
internet3:race5		-0.083 (0.552)	-0.777 (0.554)	0.506 (0.777)
internet4:race5		-0.094 (0.643)	-1.101 (1.041)	0.849 (0.910)
internet5:race5		-4.031*** (0.987)	-2.462*** (0.866)	-0.287 (1.573)
internet2:party2	0.309 (0.227)	0.500*** (0.090)	0.469*** (0.092)	-0.543*** (0.145)
internet3:party2		0.757*** (0.215)	0.326 (0.215)	-0.455 (0.326)
internet4:party2		0.421 (0.337)	-0.793*** (0.304)	-0.438 (0.496)
internet5:party2		1.376** (0.604)	1.009* (0.521)	-0.259 (0.736)
Observations	9,999	9,542	9,374	3,778

Note:

*p<0.1; **p<0.05; ***p<0.01

INTERPRETATIONS / RECOMMENDATIONS

- Does more frequent internet use increase the likelihood of strong partisan affiliation among US adults?
 - **Republican** Affiliation
 - There are inconsistent effects
 - 2020 and 2023, more frequent internet usage decreases the odds of stronger Republican affiliation
 - 2021 and 2022, more frequent internet usage increases the odds of stronger Republican affiliation.
 - **Democratic** Affiliation
 - There are inconsistent effects
 - More evidence that more frequent internet usage decreases the odds of stronger Democratic affiliation.
- Future Scope
 - Investigate the effect of factors such as political events or online content consumption

INTERPRETATIONS / RECOMMENDATIONS

- Does the effect of internet use on partisan affiliation vary by age or race?
 - **Republican** Affiliation:
 - Age: Across the 4 years, there is not significant difference in Republican favorability by age
 - Race:
 - 2020: Among people who use the internet 'Several times a day', individuals of Mixed Race have 1.14 higher log-odds to have a favorable opinion of the Republican Party compared to White individuals who use the internet 'Almost constantly'.
 - 2021: Among those who use the internet 'Several times a week', being Asian or Asian-American is associated with a 2.7 lower log-odds of having a more favorable opinion of the Republican Party compared to White individuals who use the internet almost constantly.
 - 2022: Among those who use the internet 'Less often', being of Mixed Race is associated with a 11.6 higher log-odds of having a more favorable opinion of the Republican Party compared to White individuals who use the internet almost constantly.
 - 2023: Among those who use the internet 'Less often', being Asian or Asian-American is associated with a 5.2 lower log-odds of having a more favorable opinion of the Republican Party compared to White individuals who use the internet almost constantly.

INTERPRETATIONS / RECOMMENDATIONS

- Does the effect of internet use on partisan affiliation vary by age or race?
 - **Republican** Affiliation
 - **Democratic** Affiliation:
 - Across the 4 years, there is not significant difference in Democrat favorability by age
 - Race:
 - 2020: Among people who use the internet 'Several times a day', Black or African-American individuals were less likely (1.305 log odds lower) to have a favorable opinion of the Democratic Party compared to White individuals who use the internet 'Almost constantly'.
 - 2021: Among those who use the internet 'Less often', identifying as 'Or some other race' is associated with a 4.03 lower log-odds of having a more favorable opinion of the Democratic Party compared to White individuals who use the internet almost constantly.
 - 2022: Among those who use the internet 'Less often', being Asian or Asian-American is associated with a -11.25 lower log-odds of having a more favorable opinion of the Democratic Party compared to White individuals who use the internet almost constantly.
 - 2023: Among those who use the internet 'Less often', being of Mixed Race is associated with a 9.9 higher log-odds of having a more favorable opinion of the Democratic Party compared to White individuals who use the internet almost constantly.

ASSUMPTIONS

		Multicollinearity (VIF)		Autocorrelation (DW)
Ordinal Logit: rep	2020	Failed	internet ~19 internet:age ~15	Inconclusive. Cannot run DW test on ordered factor DVs.
	2021	Failed	Internet: ~7e5 internet:age ~6e5	
	2022	Failed	Internet: ~1.1e10 Internet:race ~2.7e7	
	2023	Failed	Internet: ~3.8e5 Internet:age ~2.99e5	
Ordinal Logit: dem	2020	Failed	internet ~19 internet:age ~15	
	2021	Failed	Internet: ~7e5 internet:age ~5.9e5	
	2022	Failed	Internet: ~8.5e12 Internet:race ~1.7e14	
	2023	Failed	Internet: ~1.1e10 Internet:race ~1.3e8	

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